

Adam Wei

📍 Cambridge, MA ✉ weiadam@mit.edu 🔗 adamwei.com 🎓 [Google Scholar](#)

Education

Massachusetts Institute of Technology

Sep 2023 – Present

Ph.D. Student in Computer Science – Advisor: [Prof. Russ Tedrake](#)

- My research focuses on two main directions: **1)** understanding and improving how different data sources can be used in robot imitation learning, **2)** algorithmic improvements for generative modeling in robotics with a focus on learning from out-of-distribution or low-quality data.

University of Toronto

Sep 2019 – May 2023

B.Eng. in Electrical Engineering

- **CGPA:** 3.99/4.0
- **ECE Top Student Award:** 2020, 2021, 2022

Publications & Preprints

Ambient Diffusion Policy: Imitation Learning from Suboptimal Data in Robotics [🔗](#)

Preprint 2026

Adam Wei, Nicholas Pfaff, Thomas Cohn, Arif Kerem Dayi, Constantinos Daskalakis, Giannis Daras, Russ Tedrake

Empirical Analysis of Sim-and-Real Cotraining of Diffusion Policies for Planar Pushing from Pixels [🔗](#)

IOS 2025

CoRL Workshop 2025

Adam Wei, Abhinav Agarwal, Boyuan Chen, Rohan Bosworth, Nicholas Pfaff, Russ Tedrake

How Well do Diffusion Policies Learn Kinematic Constraint Manifolds? Best Poster Award [🔗](#)

Under Review 2025

IOS Workshop 2025

Lexi Foland, Thomas Cohn, Adam Wei, Nicholas Pfaff, Boyuan Chen, Russ Tedrake

Consensus Complementarity Control for Multi-Contact MPC Best Paper Award [🔗](#)

IEEE T-RO Journal 2024

Alp Aydinoglu, Adam Wei, Wei-Cheng Huang, Michael Posa

Framework and Software for Real-Time Multi-Contact Model Predictive Control

RSS 2022

Alp Aydinoglu, Adam Wei, Michael Posa

BitAllocation: A Resource Allocation Algorithm For Fixed-Point Quantization

Preprint 2021

Adam Wei, Andreas Moshovos

Talks

Imitation Learning From Suboptimal Data

[Slides](#) [🔗](#) *Physical Intelligence*

Cambridge, MA / San Francisco, CA

Jun 2026 / Jun 2026

Empirical Analysis of Sim-and-Real Cotraining

[Slides](#) [🔗](#) *IOS 2025 / CoRL Workshop 2025*

Hangzhou, China / Seoul, Korea

Oct 2025 / Sep 2025

Learning From Out-of-Distribution Data in Robotics

[Slides](#) [🔗](#) *Amazon Science Hub Robotics Research Day 2025*

Cambridge, MA

Sep 2025

Principles of Sim-and-Real Cotraining for Robot Manipulation

[Slides](#) [🔗](#) *Amazon Consumer Robotics Symposium 2025*

Sunnyvale, CA

Apr 2025

Fellowships & Awards

Best Paper Award, <i>IEEE RAS TC on Model-based Optimization for Robotics</i>	2025
Graduate Research Fellowship, <i>National Science Foundation</i>	2024
Canada Graduate Scholarships-Doctoral, <i>NSERC</i>	2024
NDSEG Honorable Mention, <i>US Department of Defense</i>	2023
Gordon R. Selmon Capstone Design Award, <i>University of Toronto</i>	2023
ECE Top Student Award, <i>University of Toronto</i>	2020, 2021, 2022
First-Year Summer Research Fellowship, <i>University of Toronto</i>	2020

Work Experience

Robotics Research Intern Physical Intelligence	<i>San Francisco, CA</i> <i>Jun 2026 – Present</i>
Robotics Ph.D. Student Massachusetts Institute of Technology – Advisor: <i>Prof. Russ Tedrake</i>	<i>Cambridge, MA</i> <i>Sep 2023 – Present</i>
My research focuses on two main directions: 1) understanding and improving how different data sources can be used in robot imitation learning, 2) algorithmic improvements for generative modeling in robotics with a focus on learning from out-of-distribution or low-quality data.	
Robotics Research Assistant University of Pennsylvania, GRASP Lab – Advisor: <i>Prof. Michael Posa</i>	<i>Philadelphia, PA</i> <i>May – Dec 2022</i>
- Developed and deployed a real-time model predictive control algorithm for multi-contact systems - Published in the IEEE T-RO journal and received a Best Paper Award (see above)	
Software Engineer Tenstorrent – <i>AI Hardware Start-up with Jim Keller</i>	<i>Toronto, Canada</i> <i>May – Aug 2021</i>
- Developed infrastructure for our chip's kernels & backend compiler using C++, Python, SQL, and Verilog - Awarded a bonus for demonstrating leadership skills and taking ownership of my projects	
Machine Learning Research Assistant University of Toronto – Advisor: <i>Prof. Andreas Moshovos</i>	<i>Toronto, Canada</i> <i>May 2020 – Dec 2021</i>
Developed a novel optimization-based quantization algorithm that achieved state-of-the-art compression of ImageNet models in linear-time (as opposed to existing algorithms that required exponential time)	

Teaching & Service

Teaching Assistant , University of Toronto ECE421: Robot Modeling and Control – <i>Prof. Lacro Pavel</i>	<i>Toronto, Canada</i> <i>Jan – Apr 2023</i>
Graduate School Application Mentor , Massachusetts Institute of Technology MIT GAAP Program	<i>Cambridge, MA</i> <i>2023 – 2024</i>
University of Toronto vs University of Waterloo: a Detailed Comparison Blog Post 	<i>2023</i>
Student Mentor , University of Toronto ECE Club Mentorship Program	<i>Toronto, Canada</i> <i>2022 – 2023</i>